Task 3 · Data Warehouse Design Justification

Fact tables and grains

Table	Grain	Measures / purpose
FACT_ECONOMY	Country × Year	GDP per capita, GDP, GDP growth, exports and imports (% GDP)
FACT_SOCIETY	Country × Year	Population, GNI per capita, life expectancy, urban population (%)
FACT_SECTOR	Country × Year × Sector	Value added (% GDP and USD), employment (%)

Why three separate facts?

Economic performance and societal wellbeing are different analytical subject areas, so they are stored in separate typed fact tables even though both have a country–year grain. Sector indicators have a finer country–year–sector grain. Combining all three in one table would repeat country-level measures across sectors and risk incorrect aggregation.

Why these dimensions?

DIM_COUNTRY and DIM_TIME are conformed dimensions shared by the three facts, enabling comparisons by country group, country, decade and year. Their primary keys are generated surrogate keys; ISO3 code and year remain unique business identifiers. DIM_SECTOR applies only to FACT_SECTOR because the other two facts are not measured by sector. Its hierarchy is Primary → Agriculture; Secondary → Industry (Manufacturing); Tertiary → Services.

Why an events bridge?

BRIDGE_EVENT holds one row per historical event and contains foreign keys to DIM_COUNTRY and DIM_TIME, as well as its event description, category and economic impact. This supports multiple events in the same country and year without introducing event text into the numerical facts. It also makes event ETL straightforward from the staging events table. Event rows must be retrieved independently of fact aggregation (or aggregated first) to avoid multiplying measures when multiple events share a country–year.